CprE 2880 Final Project Proposal

(SE-PG#5 AKA Coolest Team Like Ice Cream)

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Problem Statement

Space is often referred to as the final frontier; however, Earth's own oceans remain largely unexplored. Less than 5% of the ocean has been explored, and only 27.3% has been mapped with high-resolution data. [6] Vast portions of the ocean floor remain uncharted, which limits our understanding of marine ecosystems, geological formations, and the distribution of resources. Some of the challenges researchers have faced in mapping the sea floor include the sheer size of the ocean and extreme conditions, such as intense water pressure, cold temperatures, and constant darkness. [10]

The goal of this project is to design and implement a Cybot-based ocean floor exploration robot capable of performing seafloor mapping in a simulated environment. Using infrared (IR) and ultrasonic (Ping) sensors, the robot will identify obstacles and terrain features such as "holes", "reefs", or "rocks" within a set area around the Cybot. The Cybot will then communicate mapping and navigation data wirelessly via UART to a Wi-Fi host computer for visualization and analysis.

Main Project Objectives:

- Accurately map and display a section of seafloor (the test field) using sensors aboard the Cybot.
- Detect, analyze, and avoid a variety of obstacles, including holes, rocks/debris (short objects), and coral structures (tall objects).
- Efficiently navigate the test field to minimize total mapping time and reduce overall power usage.
- Transmit real-time position and sensor data, allowing the group to map the test field for data collection and monitoring purposes.

Application Narrative

With rapid technological advancement and growing interest in deep-sea research, the need for affordable, modular, and accessible autonomous systems has become crucial. Professional organizations such as the Monterey Bay Aquarium Research Institute (MBARI) have developed compact underwater robots that make deep-sea exploration more accessible to scientists and students alike. [4]

Our Cybot prototype serves as a surface-level analog to these underwater systems. It will simulate the exploration of an “ocean floor” environment in a controlled lab setting, using IR, ultrasonic, and cliff sensors to detect and map terrain variations. Within a practical use setting, the bot would be delivered to the proposed area of exploration by a sea vessel and then dropped in the center of the exploration area. Once the bot is placed at the bottom of the ocean, through Wi-Fi UART communication, the Cybot will relay data for data collection, remote navigation, and visualization on a PC.

The broader purpose of this project is to demonstrate how embedded systems can be applied to marine robotics, creating pathways for future enhancements, such as multiple-robot coordination, adaptive path algorithms, and environmental sensing/mapping in less accessible locations.

While this project draws inspiration from real AUVs, the proposed Cybot implementation intentionally abstracts the ocean environment into a controlled terrestrial, lab-setting-based test field. This abstraction ensures we keep our project goals within the scope of the system constraints—such as the absence of allowed camera-based sensors—while still preserving the core challenges of remote navigation, obstacle detection, boundary adherence, and data relay, which are inevitable in a real-world marine exploration environment.

User Research

User types/groups:

- Ocean biologist/Scientist attempting to better research the ocean.
- Marine life activists who are dedicated to minimizing ecological impacts.

- Commercial surveyor/inspector looking to ensure commercial activities are carried out in legal, professional, and ethical manners.

Empathy Maps

To design a meaningful and effective ocean exploration robot, we analyzed user groups that would benefit most from autonomous seafloor mapping. Each group was defined using literature on AUV user needs, operational constraints, and field research practices. The following empathy maps reflect this goal and identify the needs of potential user groups:

Marine Biologists and Scientists

Do	Think
❖ Plans and conducts lengthy survey missions [3]. ❖ Relies on remotely collected sensor data rather than direction observation [3]. ❖ Reviews terrain maps and sensor data during and following missions for analysis [3]. ❖ Works within limited research budgets and observes deployment window limitations [3].	❖ "If the data is not reliable, the mission may need to be repeated [3]." ❖ "Human access is risky or impossible for many target environments [2]." ❖ "Consistent, accurate coverage matters more than outstanding resolution [1]." ❖ "We may need systems that operate with minimal intervention, but drivers must be able to reliably interpret data [1]."
Say	**Feel**
❖ "We cannot afford to lose data due to sensor failure [1]." ❖ "Repeatability is critical for validating the test environments [1]." ❖ "The quality and accuracy of the sensor data determine the quality of our results [3]."	❖ Concerned about the reliability of sensor data and the inability to debug while in use [1]. ❖ Relieved that remote systems reduce human risk [2]. ❖ Motivated by scientific discovery and a high degree of environmental stewardship. Feels proud that this technology enables the advancement of the marine research frontier and the

	preservation of ecosystems [1].

Marine Activists and Surveyors

Do	Think
❖ Monitors the environmental impact of exploration or commercial activity [5]. ❖ Reviews survey data to assess ecosystem health over time [10]. ❖ Advocates for responsible and non-invasive research practices [2]. ❖ Works with limited funding and access to specialized equipment [9].	❖ "Exploration should not harm the ecosystems it studies [10]." ❖ "Data collection must justify its environmental impact [10]." ❖ "Affordable tools could expand conservation efforts [10]." ❖ "Monitoring needs to be continuous, not a one-time thing [9]."
Say	**Feel**
❖ "Non-invasive sensing is essential [10]." ❖ "We need evidence to hold organizations accountable [4]." ❖ "High-cost tools exclude smaller conservation groups [9]."	❖ Protective of marine ecosystems and habitats [10]. ❖ Skeptical of commercial exploration motives [9]. ❖ Frustrated by the lack of accessible monitoring tools [9]. ❖ Empowered when data supports conservation decisions [10].

Point Of View Statements

- A marine biologist requires consistent and repeatable access to environmental terrain information, as direct human observation is often unsafe or impossible in deep or remote ocean regions.
- A commercial surveyor requires a reliable and repeatable mapping tool to ensure that underwater infrastructure is inspected consistently and efficiently.
- A marine life activist needs a way to monitor marine habitats passively because they want to track changes in underwater ecosystems without disturbing them.
- A marine life activist needs an affordable monitoring tool because they want to support conservation in areas with limited research funding.

- A marine life activist needs a way to observe environmental changes without disturbing the habitat because they value non-invasive research practices.
- A commercial surveyor needs a way to verify environmental or structural changes over time, as they are accountable for the quality of maintenance.
- A commercial surveyor requires a tool that can detect and avoid unexpected obstacles, as they must prevent equipment collisions in confined spaces.

Unifying Point Of View Statement

All primary user groups—marine biologists, environmental activists, and commercial surveyors—need an accurate, sensor-driven robotic system capable of gathering, transmitting, and visualizing environmental data because they all share the goal of obtaining reliable, repeatable insights from difficult-to-access underwater environments while minimizing human risk, operational costs, and ecological disturbances.

Function Requirements and Mapping:

Large Functional Requirements

Base Cybot Capabilities	Default Usage	Project Usage	Implementation
Open Interface	General Cybot Movement Position Tracking Playing Sounds	oi_t oi_setWheels Encoder values	We will use it to move and track our bot across the seafloor environment.
Interrupts	Communication	Emergency quit function in the movement cases.	We used the interrupt function to allow the operator to override any issues or unexpected situations that may arise, to leave the code.

Wifi-UART	Cybot Communications	Putty get byte Putty send byte Putty send str	Communication Between the user and the CyBot, this enables the CyBot to relay data back to the user, allowing for manual adjustments if needed.
ADC	Infrared Sensor Cliff Sensor	Adc init and read	Needed to read IR and cliff sensors; may also need to read additional sensors, such as a gyroscope for 3D mapping or an altimeter for depth.
Input Capture	Ping Sensor	Used to map objects	This is used to map objects in front of the bot to create a display of the objects in the given area
PWM	Servo Motor	Used for communication and servo movement	Sensor communication readings and initialization.

Lotus Blossom Diagram

The following Lotus Diagram was used to expand the central goal of a CyBot-based seafloor mapping robot. It was broken down into its primary functional areas, and each petal informed one or more of the large functional requirements, aiding traceability from user needs to the system implementations.

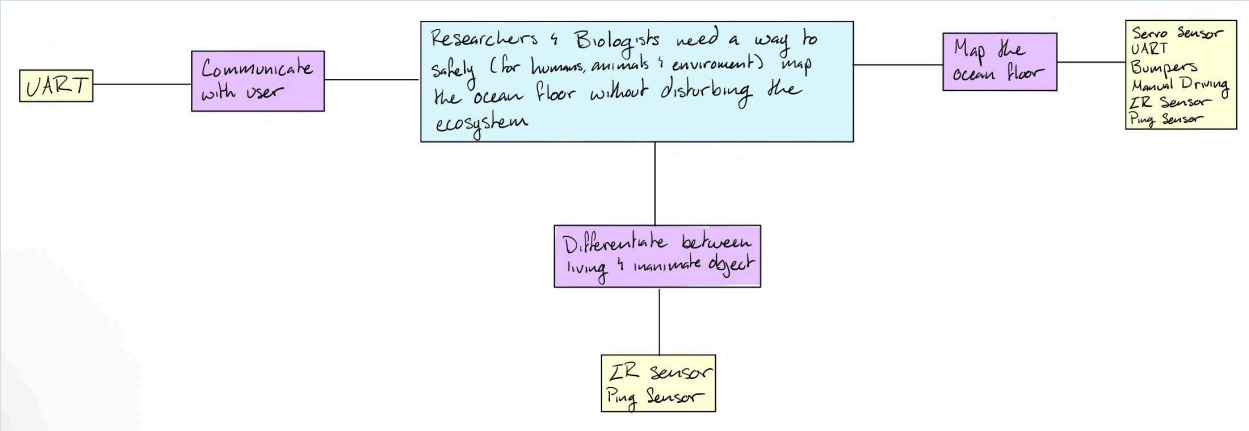

Figure 1: The project team's mapping of CyBot capabilities in a lotus diagram format

Testing / Sketches

User Sketch

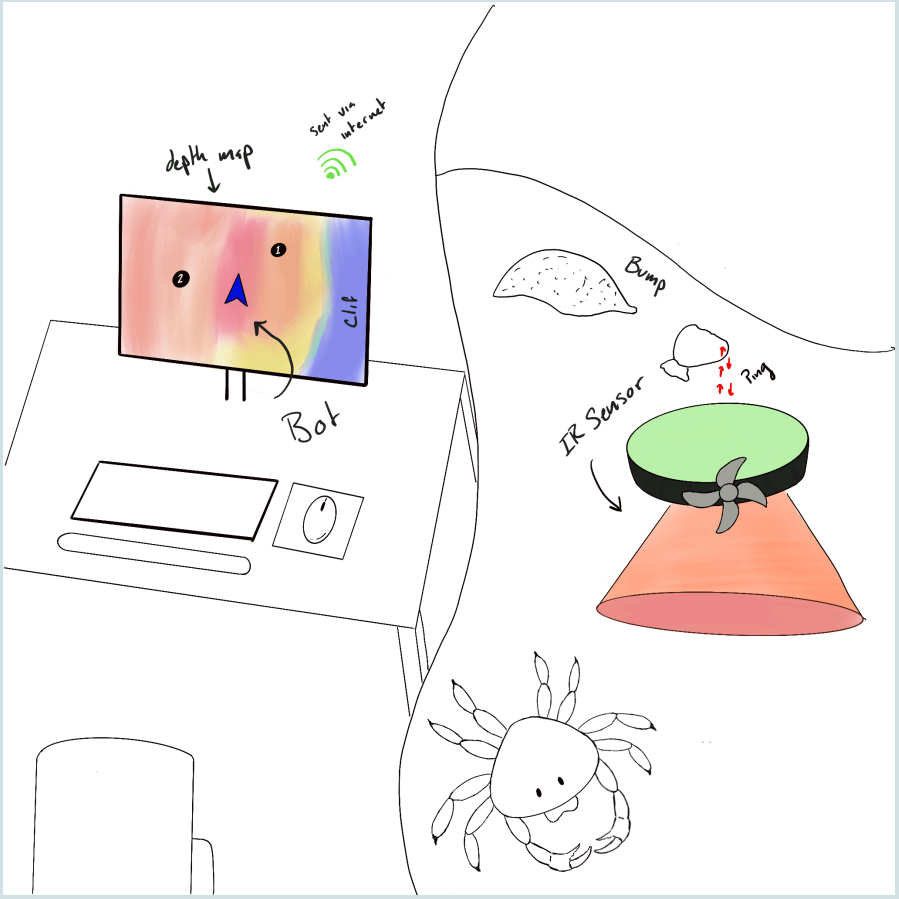

Figure 2: The project team's user-centered sketch

This user-centered sketch illustrates how the CyBot system is deployed within the context of a simulated ocean floor exploration mission. Here, the CyBot represents the underwater vehicle that, through remote operation, effectively and accurately maps the target area. The human driver that would make use of the computer setup represents a marine researcher or surveyor interacting with the system through a robust GUI.

As shown, the CyBot navigates the exploration area while collecting sensor data related to obstacles, terrain changes, and boundary regions. This data is transmitted wirelessly to the host computer, where it is visualized for the operator. The driver utilizes this real-time feedback to inform navigation decisions, establish safe paths, and ensure comprehensive coverage of the target area.

At a high level, this sketch demonstrates the project's ability to meet core functional requirements. Combining information from the IR, Ping, bump, and cliff sensors, coupled with

the CyBot's movement and user interface interactions, the bot effectively achieves the project's main objectives.

Technical Diagram

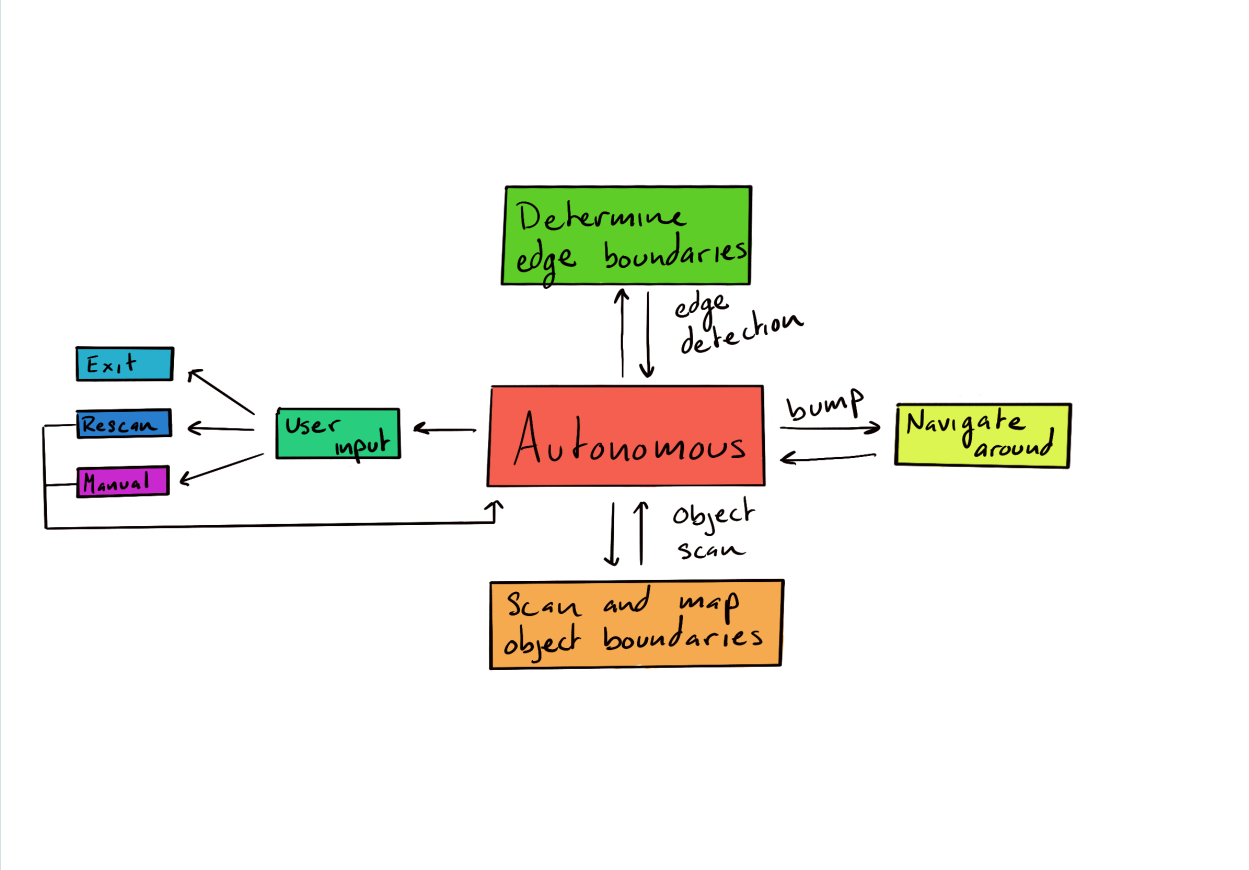

Figure 3: The project team's technical sketch illustrating subsystem interactions

This sketch presents the technical architecture of the CyBot system, highlighting how both hardware and software components interact to support the project's functionality. Infrared sensors are connected through the analog-to-digital converter (ADC) to detect nearby obstacles and terrain features. An ultrasonic ping sensor, accessed through input capture, provides distance measurements for object mapping. Cliff sensors enable the detection of holes or sudden drop-offs. Motor control is achieved through the CyBot open interface, allowing the driver to implement precise wheel movement commands.

Many of these features also demonstrate bidirectional communication between the CyBot and the host computer, which is achieved through Wi-Fi UART. This enables real-time transmission of positional and sensor data to the driver's GUI.

Test Field

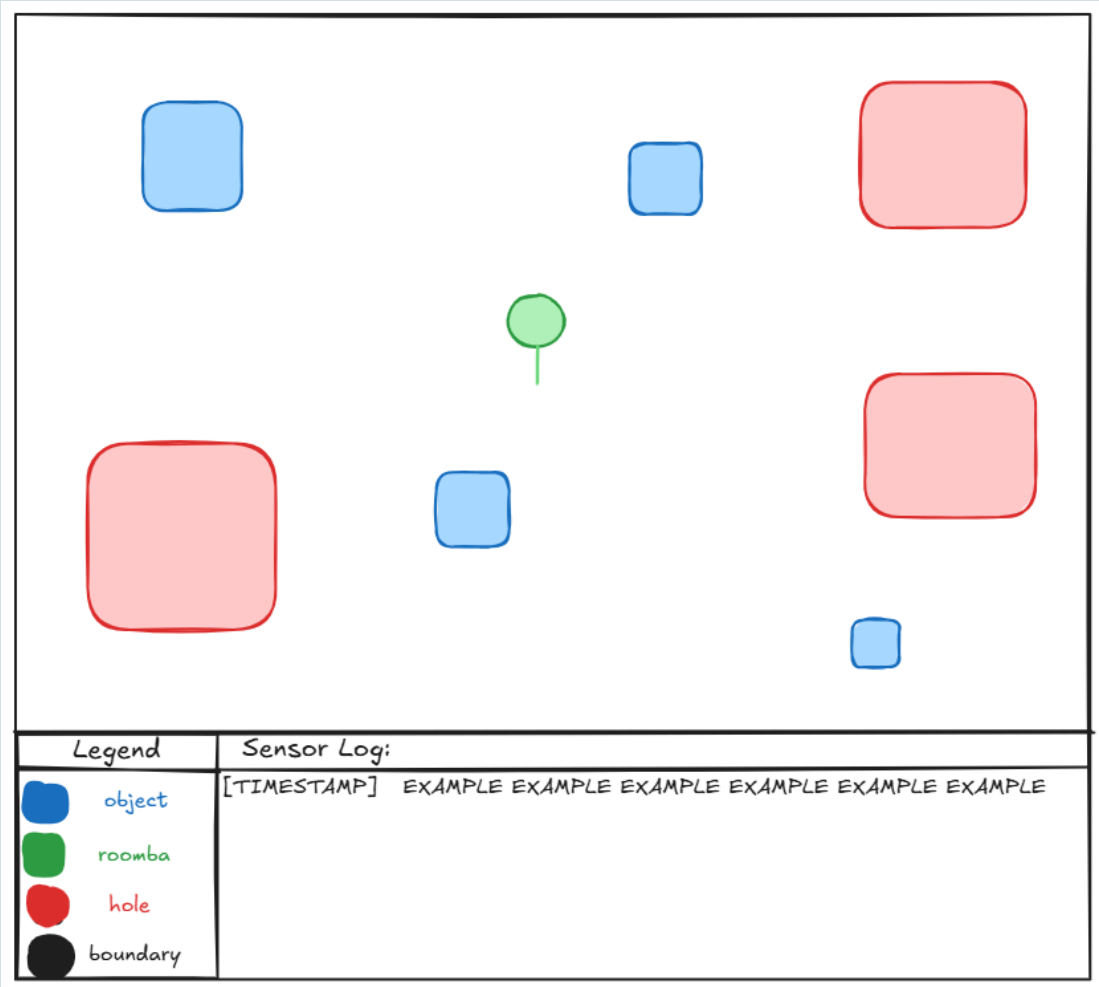

Figure 4: An example of a mock test environment for the CyBot

At a high level, this sketch illustrates how a mock test field might be setup and used to evaluate the CyBot's performance in a controlled environment. A variety of obstacle types are employed to mimic real-world variations in terrain, landscape features, and potential navigation hazards.

During testing, the CyBot is remotely navigated through this environment while continuously scanning for obstacles, holes, and boundaries. Sensor detections are transmitted to the GUI, allowing the operator to visualize the environment and guide the CyBot along safe and efficient routes. Each test field element corresponds directly to a functional requirement, ensuring that movement, object detection, boundary adherence, and task completion can be evaluated in a repeatable and measurable manner.

GUI Mini Map-Dashboards

The figure below represents the various sources of sensor data that were combined into the comprehensive display illustrated in later versions of the GUI, which can be found in later figures. It demonstrates how each of the individual sensor data types was recorded and displayed.

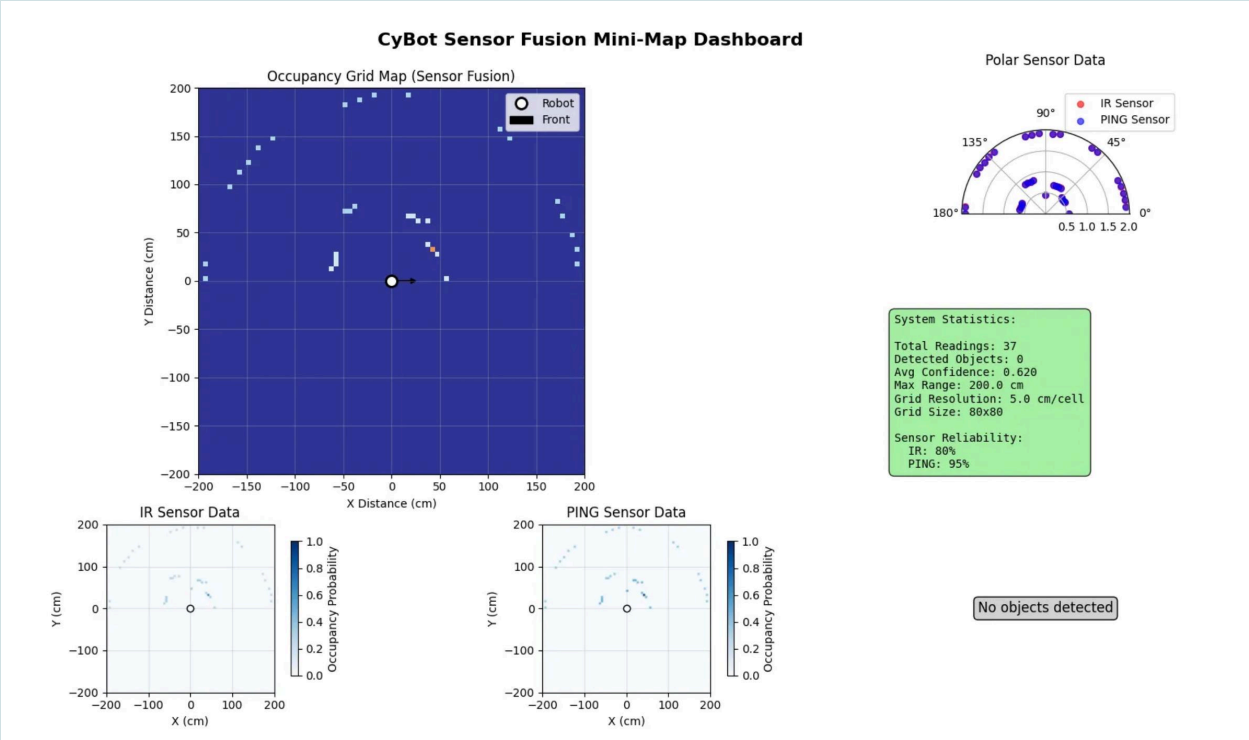

Figure 5: Demonstrating the first stages of the project's GUI development

The figure below demonstrates how the GUI evolved into a comprehensive display. The driver gained the ability to manually press movement commands, while still preserving the wasd keyboard input capabilities incorporated into the C code.

The driver also gained the ability to toggle sensor data on and off, facilitating management of the visual clutter. Among other functions, the GUI at this point was now able to display sensor data and record it in the communication log at the bottom of the figure, as well as the ability to perform each type of scan individually or collectively in sequence.

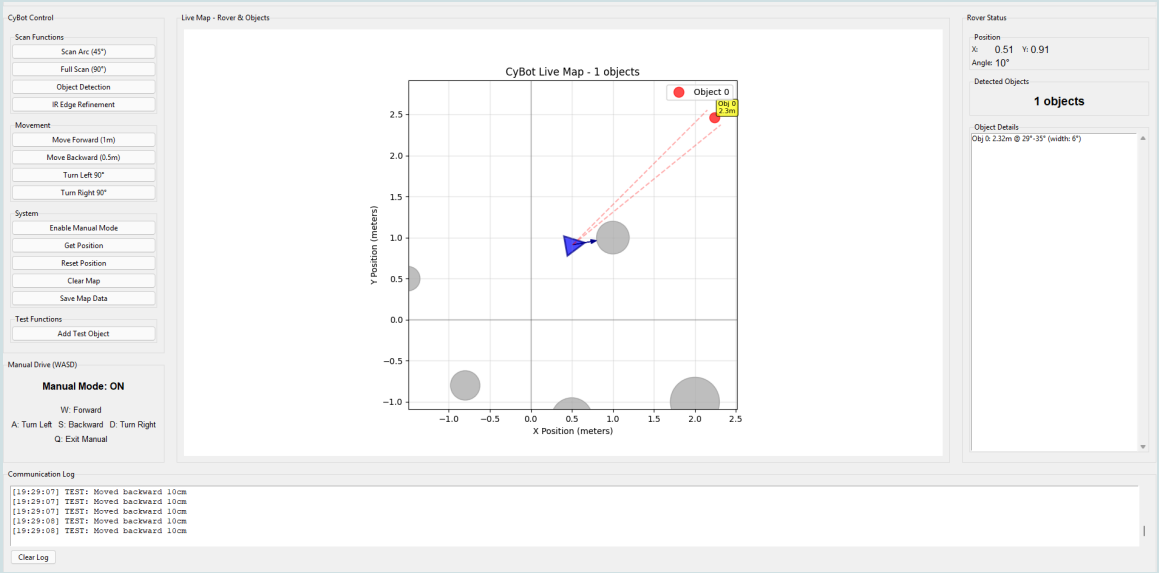

Figure 6: The final stages of GUI development, demonstrating the features and displays.

The final sketch, found below, demonstrates several of the minimap features allowing for the driver to analyze the visual data. Here, the object menu is enabled to show both the location, type, and distance of the detected objects. Additional display menus for object types, such as cliffs/holes and boundaries, are not shown here but are also available to the driver at their discretion.

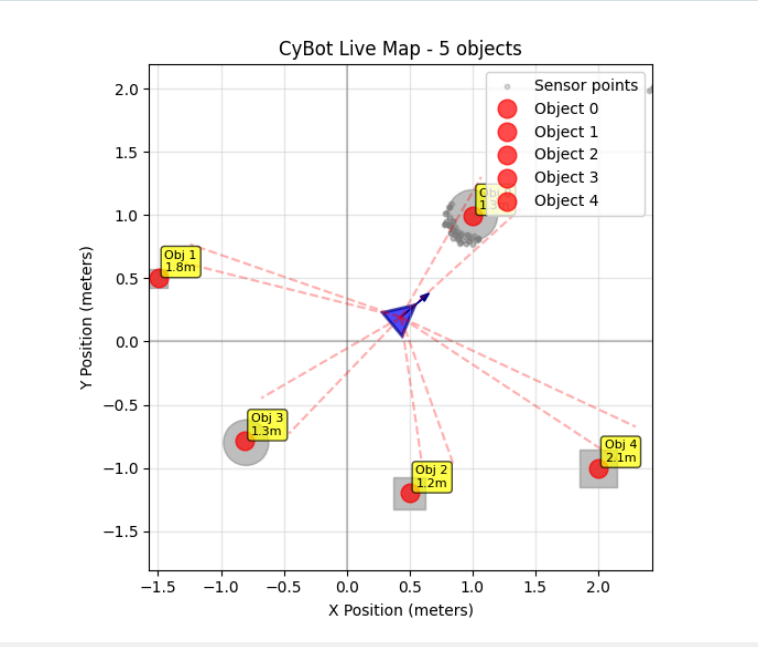

Figure 7: An image demonstrating the types of mapping menus available to the driver during and after mapping a target area.

Collectively, these sketches demonstrate how the project's large functional requirements are realized through specific CyBot platform capabilities and validated within a representative test environment. This robust GUI facilitated many of the project's main objectives and allowed the project team to successfully and remotely navigate the CyBot through test environments.

Demo Narrative:

During the team's final demonstration, the CyBot will be placed at a random starting location, positioned approximately near the center of the mock environment. Once initialized, the bot will perform an initial sensor scan to establish baseline positional and environmental data.

Throughout the demonstration, the CyBot will be remotely operated by a driver using commands issued from a host computer via Wi-Fi UART communication. As the robot navigates the test field, infrared, ultrasonic, and cliff sensors will continuously collect data to identify obstacles, holes, and boundary regions. This sensor and positional data will be transmitted in real time to the host computer, where the graphical user interface (GUI) will update continuously to reflect the detected environment. Repeated scans of the same objects may occur, but this only reinforces the accuracy of the GUI map, as intended.

The GUI will assist the driver by visualizing obstacle locations, boundary limits, and previously scanned regions, enabling informed navigation decisions. When necessary, collision detection and boundary detection alerts—in the form of emergency backup procedures and audio cues available to the driver—will support the driver by warning of hazardous conditions, helping to further reduce the likelihood of any potential operator error and/or damage to the CyBot itself.

Using this combination of remote control and sensor-assisted navigation, the CyBot will efficiently survey the test field while minimizing the traversal of both unsafe and repeated routes. Upon successful completion of the mapping tasks, the driver will direct the CyBot back to its initial deployment location for retrieval, where a final audio cue will signal the completion and conclusion of the demonstration.

Collaboration

Team Member	***Contributions***
Wil	❖ Identified user groups and contributed to empathy map statements. ❖ Collected and assimilated the code of both groups and refined them into a single, cohesive whole. ❖ Implemented the basic functional requirements of movement, position and scan data. ❖ Drove for the demos.
David	❖ Contributed significantly to establishing the problem statement. ❖ Helped build and flesh out empathy maps, point-of-view statements, and narrative statements throughout the project. ❖ Implemented the main interrupt capabilities in the form of safety measures.
Alex	❖ Created the entirety of the GUI, providing a polished, refined system to make use of the underlying project features. ❖ Worked with other team members to integrate the GUI into the main body of code.
Ivy	❖ Helped to create the test field and lotus sketches, enabling the group to establish GUI display specifications. ❖ Ensured empathy map statements were properly cited and in scope of the project goals.
Ben	❖ Primary researcher responsible for gathering data on user types/groups and identifying their wants, needs, and concerns. ❖ Responsible for source collection,

	documentation, and citation. ❖ Implemented the cliff sensor into the project's movement and scan procedures to ensure reliable/accurate hole/cliff detection. ❖ Ensured that the proposal and documentation were robust and appropriately formatted.

Documenting how collaborative efforts shaped POV Statements:

The development of the final Point of View (POV) statement was the result of extensive collaboration between all team members, guided by both user research findings and iterative group discussion. Each member contributed distinct insights derived from empathy mapping, user persona analysis, and examination of secondary sources on potential AUV (Autonomous Underwater Vehicle) user groups.

- Ben: Focused on the marine biologist user group, gathering data from academic sources such as MBARI's research on autonomous underwater robots [1]. This input emphasized the importance of safe and repeatable data collection, forming the foundation of the "autonomous mapping and terrain visualization" need statements.

- Alex: Applied technical constraints from the Cybot hardware to evaluate which user needs were realistically achievable in our mock environment. This team member's inputs refined the POV statements to ensure they remained within the robot's sensing, control, and computational limits.

- Ivy: Contributed insights from environmentalist and activist perspectives, identifying the importance of low environmental disturbance and affordability. These considerations led to the inclusion of the themes "non-invasive sensing" and "low-cost accessibility" in the final statement.

- David: Focused on commercial surveyor needs, emphasizing repeatability, reliability, and inspection-oriented behavior. Their observations helped balance scientific and industrial priorities in the unifying statement.

- Wil: Analyzed the various users' needs to better understand them and their interconnections. Also contributed to their respective empathy maps, and tailored the project to best accommodate both the researchers and activists who benefit from it.

Group Discussion and Synthesis

Through discussion, the team compared overlapping needs across all user categories. The group found that all users shared three priorities—autonomy, accuracy, and minimal disturbance—which directly shaped the final unified POV statement. Each team member reviewed the draft versions to ensure the statement reflected both user diversity and technical feasibility.

This collaborative synthesis process transformed individual user insights into a coherent statement that accurately captures the shared motivations of potential user groups and the technical possibilities within the Cybot platform.

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